

2 **(a)** Show that the vector $\mathbf{i} + 4\mathbf{j} + 2\mathbf{k}$ is parallel to the plane $2x + y - 3z = 10$. **[3]**

(b) Determine the acute angle between the planes $2x + y - 3z = 10$ and $x - y - 3z = 3$. **[4]**